

Matplotlib + EDA + Data Preprocessing — ML Cheat Sheet

Matplotlib

Function / Method	Syntax	Simple explanation
plot()	plt.plot(x, y)	Draws a line graph.
scatter()	plt.scatter(x, y)	Shows the relationship between two numerical values.
bar()	plt.bar(categories, values)	Compares values across categories.
hist()	plt.hist(data, bins=4)	Shows the distribution of numerical data.
boxplot()	plt.boxplot(data)	Shows median, spread, and possible outliers.
title()	plt.title("Age vs Salary")	Adds a graph title.
xlabel()	plt.xlabel("Age")	Labels the X-axis.
ylabel()	plt.ylabel("Salary")	Labels the Y-axis.
grid()	plt.grid()	Adds grid lines.
legend()	plt.legend()	Shows labels for multiple plots.
subplot()	plt.subplot(1, 2, 1)	Places multiple plots in one figure.
xlim()	plt.xlim(15, 45)	Sets the X-axis range.
ylim()	plt.ylim(10000, 70000)	Sets the Y-axis range.
figure()	plt.figure(figsize=(10, 6))	Sets graph size.
savefig()	plt.savefig("graph.png")	Saves a graph as an image.
show()	plt.show()	Displays the graph.

EDA

Function / Method	Syntax	Simple explanation
head()	df.head()	Shows the first 5 rows.
shape	df.shape	Shows rows and columns.
dtypes	df.dtypes	Shows data types.
describe()	df.describe()	Shows count, mean, standard deviation, minimum, quartiles, and maximum.
isnull()	df.isnull().sum()	Counts missing values in each column.
duplicated()	df.duplicated().sum()	Counts duplicate rows.
unique()	df["city"].unique()	Shows unique category values.
value_counts()	df["city"].value_counts()	Counts each category.
groupby()	df.groupby("city")["salary"].mean()	Calculates a statistic for each group.
corr()	df["age"].corr(df["salary"])	Measures the linear relationship between two columns.
Correlation matrix	df.corr(numeric_only=True)	Shows correlations between numerical columns.
quantile()	df["salary"].quantile(0.25)	Finds a percentile such as Q1 or Q3.
IQR	iqr = q3 - q1	Measures the spread of the middle 50%.
Outlier bounds	lower=q1-1.5*iqr; upper=q3+1.5*iqr	Defines usual IQR limits for possible outliers.

Data Preprocessing

Function / Method	Syntax	Simple explanation
mean()	df["age"].mean()	Calculates the mean of available values.
fillna()	df["age"] = df["age"].fillna(avg)	Fills missing values.
get_dummies()	pd.get_dummies(df, columns=["city"], dtype=int)	Converts categories into numerical one-hot columns.
X and y	X = df[["age", "experience"]]; y = df["salary"]	Separates features X from target y.
train_test_split()	train_test_split(X, y, test_size=0.2, random_state=42)	Splits data into training and testing sets.
StandardScaler()	scaler = StandardScaler()	Creates a standard feature scaler.
fit_transform()	X_train_scaled = scaler.fit_transform(X_train)	Learns scaling from training data and transforms it.
transform()	X_test_scaled = scaler.transform(X_test)	Transforms test data using the training scaler.
Data leakage rule	Fit preprocessing only on training data.	Prevents test-set information from entering training.